

CSS Box Model

A Complete Visual Guide to Understanding the CSS Box Model

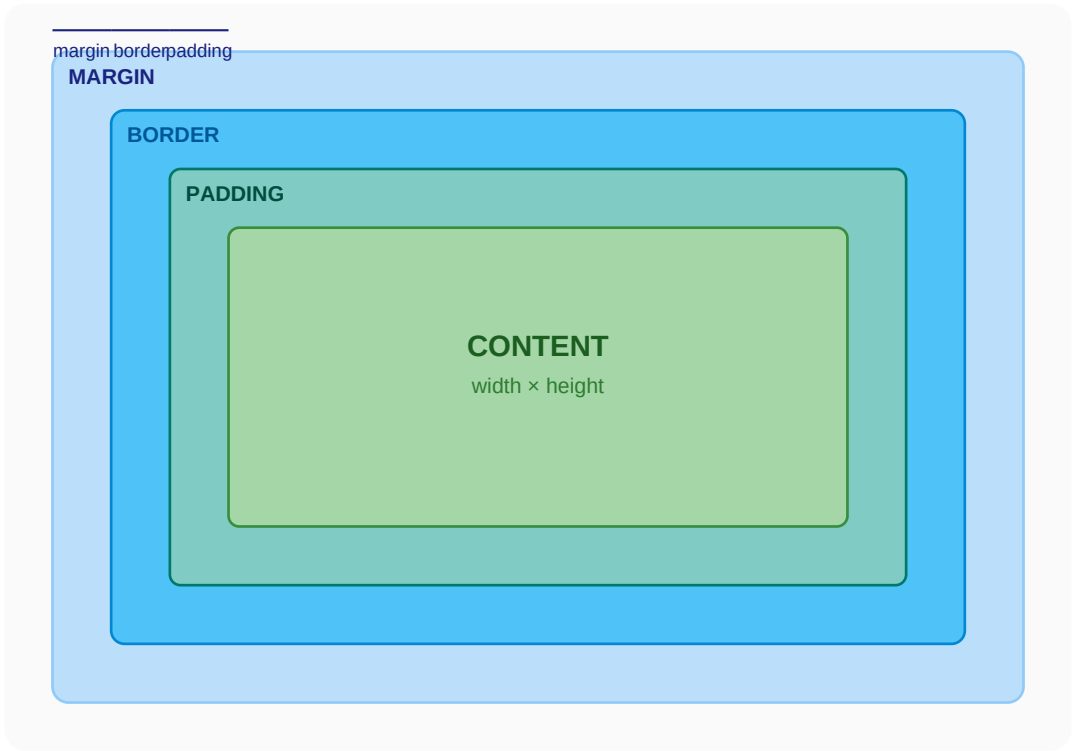
What is the CSS Box Model?

Every HTML element in a webpage is rendered as a rectangular box. The **CSS Box Model** describes how this box is structured and how space is calculated around it. Understanding the box model is fundamental to mastering CSS layout — it controls how elements take up space, interact with neighbours, and respond to sizing rules.

The box model consists of four layers (from innermost to outermost):

Layer	Description	CSS Property
Content	The actual content (text, images, etc.)	width, height
Padding	Space between content and border	padding
Border	A line surrounding the padding	border
Margin	Space outside the border	margin

Visual Diagram



1. Content

The **content area** is the innermost layer where the actual content (text, images, video, etc.) lives. Its dimensions are set using the **width** and **height** CSS properties.

```
div { width: 300px; height: 150px; }
```

2. Padding

Padding is the transparent space between the content area and the border. It expands the clickable/visible area of an element and takes the element's background color.

```
/* Shorthand: top right bottom left */ div { padding: 10px 20px 10px 20px; /* all four sides */ padding: 10px 20px; /* top/bottom left/right */ padding: 10px; /* all sides equal */ /* Individual sides */ padding-top: 10px; padding-right: 20px; padding-bottom: 10px; padding-left: 20px; }
```

■ *Note: Padding cannot have negative values. It always adds space inside the element.*

3. Border

The **border** wraps around the padding and content. It can be styled with a width, style, and color. Borders add to the total size of the element (in the default box model).

```
/* Shorthand: width style color */ div { border: 2px solid #1565c0; /* Individual properties */ border-width: 2px; border-style: solid | dashed | dotted | double | none; border-color: #1565c0; /* Individual sides */ border-top: 3px dashed red; border-radius: 8px; /* rounded corners */ }
```

4. Margin

Margin is the outermost transparent space around the border. It creates separation between elements. Unlike padding, margins can have negative values and can collapse between adjacent elements.

```
div { margin: 20px; /* all sides */ margin: 10px auto; /* centered horizontally */ margin-top: -10px; /* negative margin is allowed */ /* Individual sides */ margin-top: 10px; margin-right: 20px; margin-bottom: 10px; margin-left: 20px; }
```

■ *Tip: Use margin: 0 auto to horizontally center a block element that has a defined width.*

box-sizing Property

The **box-sizing** property changes how the total width and height of an element is calculated. It is one of the most important CSS properties for predictable layouts.

content-box (default)	border-box (recommended)
Total width = width + padding + border If width: 300px, padding: 20px, border: 2px Total = 300 + 40 + 4 = 344px	Total width = width (padding & border included) If width: 300px, padding: 20px, border: 2px Total = 300px (padding/border fit inside)
box-sizing: content-box;	box-sizing: border-box;

■ *Best Practice: Apply border-box globally to all elements for more predictable sizing:*

```
*, *::before, *::after { box-sizing: border-box; }
```

Total Size Calculation

Model	Total Width Formula	Total Height Formula
content-box	width + padding-left + padding-right + border-left + border-right	height + padding-top + padding-bottom + border-top + border-bottom
border-box	width (padding & border are included)	height (padding & border are included)

Margin Collapse

When two vertical margins meet, they **collapse** into a single margin equal to the larger of the two. This only happens with vertical (top/bottom) margins, not horizontal ones.

```
/* Example: margin collapse */ h1 { margin-bottom: 30px; } /* bottom margin = 30px
*/ p { margin-top: 20px; } /* top margin = 20px */ /* Actual gap between h1 and p =
30px (not 50px) */ /* The larger margin wins! */ /* To prevent collapse: use
padding, border, or overflow: hidden */
```

Quick Reference Cheatsheet

Property	Values / Example	Notes
width / height	300px 50% auto fit-content	Sets content area size
padding	10px 10px 20px 5px 10px 15px 20px	Transparent, takes bg color
border	2px solid #333 dashed dotted none	Adds to element size (content-box)
margin	20px auto -10px 10px auto	Can be negative; can collapse
box-sizing	content-box border-box	border-box recommended
outline	2px solid blue	Like border but outside; no space taken
overflow	visible hidden scroll auto	Controls clipped content